

A Three-Party Hierarchical Game for Physical Layer Security Aware Wireless Communications With Dynamic Trilateral Coalitions

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Abstract—In this paper, a novel hierarchical game framework for physical layer security (PLS) aware wireless communications with dynamic trilateral coalitions is studied. In the considered system, legitimate users (LUs) aim to transmit secret data to associated base stations (BSs) via uplink communications under the threat of eavesdroppers (EVs), while there also exist jammers (JAs) which may choose to form coalitions with either LUs for increasing their secrecy transmission rates or EVs for increasing their eavesdropping rates in exchange for potential rewards. Different from the existing work, we explore such complicated while dynamic coalition relationships under uncertainties of wireless systems (e.g., time-varying channel conditions), and formulate a hierarchical game integrated with a dynamic trilateral coalition formation game to model strategic interactions among LUs, JAs and EVs. Particularly, we first analyze stability requirements of the trilateral coalitions and propose a hedonic coalition selection and formation algorithm for reaching a stable coalition partition in each time slot. In addition, we propose a deep reinforcement learning (DRL) based solution, which can achieve the equilibrium with long-term performance guarantees for the hierarchical game running over multiple time slots with dynamic evolutions. Simulations evaluate the proposed solution and show its superiority over its counterparts.

Index Terms—Physical layer security, hierarchical game, dynamic trilateral coalitions, deep reinforcement learning.

I. INTRODUCTION

THE security of wireless communications has become a major concern in recent years due to the increasing prevalence of open signal propagation environments. Traditional

approaches to secure wireless communications, such as cryptography-based methods implemented in the network or application layers, suffer from the high computational complexity and resource consumptions associated with message encryptions and decryptions, as well as key managements [2], [3]. As a powerful supplement or even an alternative, physical layer security (PLS), due to its capability of exploiting the intrinsic physical properties of wireless channels against eavesdropping attacks, has been proposed as an effective way to protect secure transmissions in various wireless applications with high confidentiality requirements, e.g., autonomous driving, remote surgery, and smart manufacturing [2], [3].

Although PLS has been widely studied in different aspects [4], most existing works did not fully explore the potential selfishness of all three parties in PLS, i.e., legitimate users (LUs), eavesdroppers (EVs) and jammers (JAs). Specifically, in practice, LUs, EVs and JAs may behave strategically and selfishly for maximizing their own utilities, while their strategies may not always be in conflict with each other but sometimes mutually beneficial. On one hand, LUs and JAs may form an alliance (or coalition) against EVs, in which LUs can offer incentives (e.g., monetary rewards or other profits) to JAs in exchange for their help to increase jamming power on EVs so as to protect secrecy. Examples include but not limited to the cooperative jamming with seller-buyer relationships between JAs and LUs [5] and the incentive jamming based secure routing in decentralized Internet of Things (IoT) [6]. On the other hand, EVs and JAs may form coalition against LUs, in which EVs can also incentivize JAs to jam LUs, forcing them to increase their data transmission power and making them vulnerable to eavesdropping. For instance, by enabling energy harvesting, EVs' friendly JAs can be encouraged to jam LUs if they receive the wireless power transferred from EVs as the compensation [7].

Despite that some pioneering works have discussed the strategic coalition formations of different parties in PLS [8], [9]. These works were mainly limited to fixed ally-adversary relationships, i.e., JAs can only unilaterally assist LUs or EVs from the beginning to the end. However, such relationships (i.e., coalition formations) among the three parties in PLS may not be pre-defined or fixed, and the resulted impact to PLS requires careful modeling and analysis. These are of great importance, but has not yet been studied in the literature (to the best of our knowledge) and are very challenging due to the following reasons.

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- 1) For the sake of own interests in PLS, besides the possible coalition formations, LUs may strategically determine their base station associations and data transmission power, EVs may choose to be active and inactive at different locations, and based on these JAs may allocate different jamming powers on different links for better rewards. This necessitates a hierarchical game with multi-dimensional strategies integrating with a trilateral coalition formation game (CFG).
- 2) Due to the uncertainties of wireless systems, e.g., time-varying channel conditions [10], all three parties' strategies in PLS mentioned above may dynamically evolve, and a long-term performance optimization requires the investigation of dynamic games. In particular, the underlying CFG is also dynamic, meaning that any two of the three parties may temporarily form coalitions and dynamically adjust, i.e., merge and split over time [11].

In this paper, we propose a novel hierarchical game framework with dynamic trilateral coalitions to characterize the strategic interactions among all three parties, i.e., LUs, JAs and EVs, in PLS-aware wireless communications. We first formulate a long-term optimization problem for respectively maximizing the utility of each party in terms of their potential strategies under system uncertainties. Then, we construct a hierarchical game to model the decision sequence and correlations of these problems. Specifically, EVs first decide to be whether active or inactive at different locations and the unit incentive for stimulating JAs' help. Then LUs decide their BS associations, transmission power allocations and determine the unit incentive for competing with EVs in acquiring JAs' help. Afterwards, JAs decide their ally, i.e., LUs or EVs, to form the coalition, and determine their jamming power allocations to help their ally. For analyzing such relationships, an underlying CFG is introduced, and the stability conditions for dynamic coalition changes over multiple time slots are given. On top of this, a novel hierarchical game framework is constructed and analyzed. After that, by proving the existence of stable coalition partition for the underlying CFG, we propose a hedonic coalition selection and formation (HCSF) algorithm for reaching stable coalition partition among LUs, JAs and EVs in each time slot. Furthermore, we theoretically show that, by applying such algorithm, the overall hierarchical game can eventually converge to an equilibrium. Finally, taking into account the dynamic nature of strategic evolutions (especially the dynamic coalition formation), we investigate the interdependence among multiple time slots, and develop a deep reinforcement learning (DRL) based solution for deriving the equilibrium solution with long-term performance guarantees.

The main contributions of this paper are summarized as follows.

- To characterize strategic interactions among LUs, JAs and EVs in PLS under system uncertainties, a hierarchical game integrated with a dynamic trilateral CFG is formulated.
- Considering the selfishness of all three parties in PLS, we propose an HCSF algorithm to derive the optimal coalition partition, and prove its feasibility. Moreover, the

equilibrium of the overall hierarchical game is defined and theoretically analyzed.

- With the objective of maximizing the long-term utilities in the given game, a DRL-based solution is proposed, which can produce the optimal strategic decisions (i.e., equilibrium) of each party in the PLS over multiple time slots in dynamic evolutions.
- Numerical simulations are conducted to evaluate the proposed framework along with the corresponding solution, and demonstrate its superiority over its counterparts.

The rest of this paper is organized as follows: Section II reviews the related work and highlights the novelties of this paper. Section III introduces the system model and problem formulation of the three-party hierarchical game for PLS-aware wireless communications. In Section IV, the underlying CFG and the overall hierarchical game are analyzed, and then an HCSF algorithm and a DRL-based solution are proposed. Simulation results are given in Section V, followed by the conclusion in Section VI.

II. RELATED WORK

Due to the great potential in improving the secrecy performance of wireless communications, the PLS has attracted increasing research attentions. For instance, Xu et al. in [12] developed a block coordinate descent based algorithm for resource management in secure ground-to-air communication assisted by LUs friendly JAs deployed on unmanned aerial vehicles (UAVs). Sun and Tian in [13] applied the stochastic geometry theory to artificial noise aided LUs' joint feedback rate and power allocation optimization for maximizing LUs' ergodic secrecy rate in multi-antenna cellular systems. Guo et al. in [14] leveraged the multi-agent DRL in making the proactive eavesdropping aided EVs' power allocations in multiple-input multiple-output (MIMO) systems. However, these papers only studied the PLS from a centralized optimization perspective without considering the selfishness of LUs, EVs and JAs.

Some researchers have started investigating the selfish behavior of JAs in particular. For example, Zhang et al. in [8] first introduced the seller-buyer relationship between JAs and LUs to ensure secrecy in cooperative jamming against untrusted relays by solving the optimal LUs' asking prices and JAs' jamming power allocation problem. In [6], Xu et al. extended the work in [8] by designing an incentive jamming scheme for stimulating JAs to aid LUs in multiple relay systems, where a Stackelberg game framework was developed to determine the optimal secure routing, LUs' payment and power allocations. In [9], the authors discussed the selfish jamming for wireless-powered communication systems, in which energy-harvesting JAs were incentivized by wireless power transferred from LUs or EVs in exchange for their help. Nevertheless, these research attempts mainly focused on the fixed ally formation that JAs can only cooperate with either LUs or EVs unilaterally. However, owing to unpredictable channel conditions in real-world wireless communication systems [15], such cooperative relationships (i.e., the coalition partition) may change over time among LUs, JAs and EVs in PLS.

Coalition formation game (CFG) as a technique for analyzing the cooperations among strategic participants, has been widely employed in wireless communication systems. For instance, in [15], Chen et al. formulated a CFG for multiple UAVs' optimal task assignment and bandwidth allocation, where cooperative UAVs can form coalitions to perform different tasks while compete for bandwidth resources. In [16], Qi et al. extended the model in [15] by allowing UAVs with different resources to simultaneously join several coalitions to complete different tasks, so that resources of some UAVs can be shared among different coalitions, resulting in overlapped coalitions. In [17], Han et al. proposed a two-stage hierarchical game with an anti-jamming CFG as a subgame for LUs's choosing allies to decrease their energy consumptions. It is worth noting that, in these papers, the coalition partitions generated by CFG-based approaches maintained static after achieving the optimal outcomes, ignoring that coalition partitions may evolve slot by slot due to system dynamics.

Some recent efforts have been devoted to studying the interdependence of coalition partitions over different time slots in CFG. In [18], a dynamic Bayesian CFG was formulated for cooperative UAV delivery planning considering the uncertainties of UAV breakdowns, in which the coalition selection of each UAV was varied according to its observation of other cooperative UAV's behaviors in previous time slots. In [19], a shapley value based CFG was employed in cooperative content caching, in which the coalition formation among fixed and mobile cache providers were dynamically updated according to time-varying shapley values. However, the solutions in these works (e.g., merge and split, hedonic shift, dynamic programming, etc.) lack the exploration of unpredictable environment states and have high computational complexities, which may not be applicable for PLS-aware wireless communication systems.

In summary, this work differs from existing literature by the following aspects: i) unlike the fixed relations employed in existing works, a dynamic trilateral CFG is exploited to formulate the dynamic ally-adversary relationships among the three parties in PLS; ii) a novel dynamic hierarchical game framework is developed to model the three parties' long-term strategic interactions in dynamic wireless network system rather than the static one; and iii) instead of the model-based algorithms with limited flexibility and adaptability to system dynamics, a DRL-based solution with strong exploration capability and training efficiency is proposed.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, the network model of a PLS-aware wireless uplink communication system involved by LUs, EVs and JAs, is first described. Then, after defining the strategies and utility functions of all three parties, i.e., LUs, EVs and JAs, we respectively formulate each party's strategy optimization problem with long-term performance guarantees. Finally, a novel hierarchical game is proposed to model the interactions among all three parties in each time slot, and consequently a repeated decision-making process over multiple time slots is constructed. For convenience, Table I lists some important notations used in this paper.

TABLE I
IMPORTANT NOTATIONS IN THIS PAPER

Symbol	Explanation
$\mathcal{N}, \mathcal{M}, \mathcal{K}, \mathcal{J}$	set of LUs, BSs, EVs or JAs
$P_n^T(t)$	transmission power of LU n
$P_j^J(t)$	jamming power of JA j
$g_{nm}(t), g_{nk}(t)$	channel gain from LU n to BS m and EV k
$g_{jm}(t), g_{jk}(t)$	channel gain from JA j to BS m and EV k
$SINR_{nm}(t)$	SINR for the link from LU n to BS m
$SINR_{nk}(t)$	SINR for the link from LU n to EV k
n_m^2, n_k^2	AWGN at BS m or EV k
$R_{nm}^T(t)$	uplink transmission rate of LU n
R_{min}^T	minimum transmission rate
$R_{nk}^E(t)$	eavesdropping rate of EV k on LU n
$R_{nm}^S(t)$	secrecy transmission rate of LU n
$x_k^{sw}(t)$	EV k 's activation decision
$x_{nm}^T(t)$	LU n 's base station association
$x_{\{LJ\}}(t), x_{\{EJ\}}(t)$	indicator of JAs' aiding LUs or EVs
$\mu_L(t), \mu_E(t)$	unit incentive from LUs or EVs
$\pi_{\mathcal{N}}(t), \pi_{\mathcal{K}}(t), \pi_{\mathcal{J}}(t)$	strategy of LUs, EVs and JAs
c_{conf}	configuration cost of JAs
c_k	activation cost of each EV
ξ_n, η_j	unit power cost of each LU n and JA j
$\mathcal{G}^H, \mathcal{G}^C$	formulated hierarchical game and CFG
G	all three parties, i.e., LUs, EVs and JAs
C_a	a single coalition a
C_{-a}	all existing coalitions except coalition a
Δ	all possible coalitions
$\mathcal{M}_i$	MDP for party i
$\mathcal{CP}(t)$	coalition partition at time slot t
$\delta_i(t)$	coalition choice of party i in time slot t
$U_{\mathcal{N}}(t), U_{\mathcal{K}}(t), U_{\mathcal{J}}(t)$	utility function of LUs, EVs or JAs
π_i^H, π_i^C	strategies of party i in $\mathcal{G}^H$ or $\mathcal{G}^C$

A. Network Model

Consider a PLS-aware wireless uplink communication system, as illustrated in Fig. 1, which consists of a group of LUs, denoted as $\mathcal{N} = \{1, 2, \dots, N\}$, aiming to transmit secret data to a set of base stations (BSs), denoted as $\mathcal{M} = \{1, 2, \dots, M\}$. For simplicity, similar to [9], we assume that each LU occupies an orthogonal channel for its uplink transmission, and thus the set of uplink channels can also be represented by $\mathcal{N}$. There exists a group of EVs, denoted as $\mathcal{K} = \{1, 2, \dots, K\}$, which may be active or inactive at different locations, and multiple JAs, denoted as $\mathcal{J} = \{1, 2, \dots, J\}$, which are capable of jamming any link in this system [17].

To depict time-varying uncertainties, a time-slotted operation framework is studied, where $t \in \{0, 1, \dots, T-1\}$ denotes the index of time slots. Due to the existence of JAs, the signal to interference plus noise ratio (SINR) of uplink transmission from any LU $n \in \mathcal{N}$ to any BS $m \in \mathcal{M}$ in time slot t can be written as

$$SINR_{nm}(t) = \frac{g_{nm}(t)P_n^T(t)}{n_m^2 + \sum_{j \in \mathcal{J}} g_{jm}(t)P_j^J(t)}, \quad (1)$$

where $g_{nm}(t)$ and $g_{jm}(t)$ stand for the instantaneous channel gains of LU n and JA j at BS m , respectively, and n_m^2 represents the additive white Gaussian noise (AWGN) at BS m , $P_n^T(t) \in \{P_{n,0}^T, P_{n,1}^T, \dots, P_{n,l-1}^T\}$ and

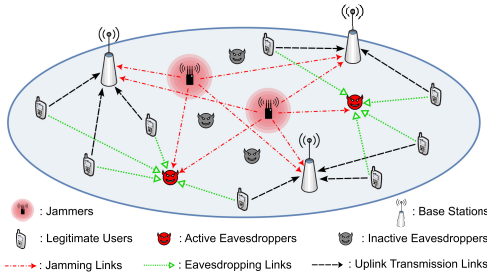


Fig. 1. Uplink communication system with LUs, EVs and JAs.

$P_j^J(t) \in \{P_{j,0}^J, P_{j,1}^J, \dots, P_{j,l-1}^J\}$ are transmission powers of LU n and JA j , respectively in time slot t , selected from finite sets with l -leveled possible values.¹ Similarly, the SINR to EV $k \in \mathcal{K}$ with regard to LU $n \in \mathcal{N}$ in time slot t can be written as

$$\text{SINR}_{nk}(t) = \frac{g_{nk}(t)P_n^T(t)}{n_k^2 + \sum_{j \in \mathcal{J}} g_{jk}(t)P_j^J(t)}, \quad (2)$$

where $g_{nk}(t)$ and $g_{jk}(t)$ stand for the instantaneous channel gains of LU n and JA j at EV k , respectively, and n_k^2 represents the AWGN at EV k . Let W denote the frequency bandwidth of each uplink channel, then the uplink transmission rate from LU $n \in \mathcal{N}$ to BS $m \in \mathcal{M}$ in time slot t can be expressed as

$$R_{nm}^T(t) = W \log(1 + \text{SINR}_{nm}(t)). \quad (3)$$

Correspondingly, the eavesdropping rate of EV $k \in \mathcal{K}$ on channel $n \in \mathcal{N}$ in time slot t can be expressed as

$$R_{nk}^E(t) = W \log(1 + \text{SINR}_{nk}(t)). \quad (4)$$

According to Wyner's definition of secrecy transmission rate in PLS [23], the secrecy transmission rate refers to an LU's transmission rate of data that can be securely delivered to its associated BS, which is calculated as the difference between LU's transmission rate and the highest eavesdropping rate on its uplink channel. Therefore, in each time slot t , the secrecy transmission rate of LU $n \in \mathcal{N}$ can be obtained as

$$R_{nm}^S(t) = [R_{nm}^T(t) - \max_{k \in \mathcal{K}} R_{nk}^E(t)]^+, \forall m \in \mathcal{M}, \quad (5)$$

where function $[x]^+ = \max(x, 0)$.

Considering that the performance gain (e.g., secrecy transmission rate and eavesdropping rate) and the latent cost (e.g., power consumption and reward payment) play important roles for practical operations of all three parties, i.e., LUs, EVs and JAs in PLS, in the following subsection, we introduce the utility function and the corresponding strategic decisions of each party in detail, and explore their interactions with resulted impacts in both short and long runs.

B. Problem Formulation

For EVs, to improve their long-term eavesdropping performance by successfully intercepting LU's confidential signals

¹This indicates that each LU or JA takes discretized transmission or jamming power in each time slot. Note that such settings are more suitable for practical implementations [20], and has been widely adopted in the literature [21], [22].

and reducing their own cost from eavesdropping, in each time slot t , they have to determine i) the activation of each EV $k \in \mathcal{K}$ at different locations, denoted as $x_k^{sw}(t) = 1$ or 0 , indicating whether EV k is active or inactive; and ii) the unit incentive² to attract JAs's cooperation, denoted as $\mu_E(t) \in [0, \mu_E^{max}]$. Note that the actual reward that JAs can get from EVs in time slot t is the product of $\mu_E(t)$ and EVs' performance gain with JAs' help in this certain slot. Such performance gain can be expressed as the difference between the gain of EVs from reducing LUs' secrecy transmission rate with JAs' help, i.e.,

$$V_{\mathcal{K}}(t) = - \sum_{k \in \mathcal{K}} x_k^{sw}(t) \sum_{n \in \mathcal{N}} \sum_{m \in \mathcal{M}} (R_{nm}^T(t) - \max_{k \in \mathcal{K}} R_{nk}^E(t)), \quad (6)$$

and the gain of EVs from reducing LUs' secrecy transmission rate without JAs' help, i.e.,

$$V_{\mathcal{K}}^0(t) = - \sum_{k \in \mathcal{K}} x_k^{sw}(t) \sum_{n \in \mathcal{N}} \sum_{m \in \mathcal{M}} W \frac{\log(1 + (g_{nm}(t)P_n^T(t)/n_m^2))}{n_m^2} - \max_{k \in \mathcal{K}} (\log(1 + (g_{nk}(t)P_n^T(t)/n_k^2))). \quad (7)$$

Then, EVs' utility function in slot t can be expressed as

$$U_{\mathcal{K}}(t) = V_{\mathcal{K}}(t) - x_{\{EJ\}}(t) \mu_E(t) [V_{\mathcal{K}}(t) - V_{\mathcal{K}}^0(t)]^+ - c_k \sum_{k \in \mathcal{K}} x_k^{sw}(t), \quad (8)$$

where $x_{\{EJ\}}(t) = 1$ or 0 indicates whether JAs choose to form an alliance with EVs or not in time slot t , and c_k is the activation cost of each single EV $k \in \mathcal{K}$ in different locations.

With EVs' strategies denoted as $\pi_{\mathcal{K}} = \{x_k^{sw}(t), \mu_E(t)\}_{\forall t}$, EV's long-term optimization problem can be formulated as

$$[\mathcal{EP}] : \arg \max_{\pi_{\mathcal{K}}} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t \in [0, T]} U_{\mathcal{K}}(t) \quad (9)$$

$$s.t., x_k^{sw}(t) \in \{0, 1\}, \forall k \in \mathcal{K}, \quad (10)$$

$$0 \leq \mu_E(t) \leq \mu_E^{max}. \quad (11)$$

For LUs, to improve their long-term secrecy transmission performance, in each time slot t , they need to determine i) the uplink transmission power allocation $P_n^T(t), \forall n \in \mathcal{N}$; ii) the BS association $x_{nm}^T(t), \forall n \in \mathcal{N}, \forall m \in \mathcal{M}$, where $x_{nm}^T(t) = 1$ or 0 signifies that LU n chooses BS m as the target receiver or not; and iii) the unit incentive $\mu_L(t) \in [0, \mu_L^{max}]$ for stimulating JAs' help. Similar to that of EVs, the actual reward that JAs can get from LUs in time slot t is the product of $\mu_L(t)$ and LUs' performance gain with JAs' help in time slot t , which can be calculated as the difference between LUs' secrecy transmission rates with JAs' help, i.e.,

$$V_{\mathcal{N}}(t) = \sum_{n \in \mathcal{N}} \sum_{m \in \mathcal{M}} x_{nm}^T(t) R_{nm}^S(t), \quad (12)$$

and such rates without JAs' help, i.e.,

$$V_{\mathcal{N}}^0(t) = \sum_{n \in \mathcal{N}} \sum_{m \in \mathcal{M}} x_{nm}^T(t) W [\log(1 + (g_{nm}(t)P_n^T(t)/n_m^2)) - \max_{k \in \mathcal{K}} \log(1 + (g_{nk}(t)P_n^T(t)/n_k^2))]^+. \quad (13)$$

²This setting has been widely employed in the literature on PLS [6], [24] and edge computing [25]. However, unlike these studies which commonly assumed that such unit incentive was fixed, we consider it as a strategic decision.

Then, LUs' utility function in time slot t can be expressed as

$$U_{\mathcal{N}}(t) = V_{\mathcal{N}}(t) - x_{\{LJ\}}(t)\mu_L(t)[V_{\mathcal{N}}(t) - V_{\mathcal{N}}^0(t)]^+ - \xi_n \sum_{n \in \mathcal{N}} P_n^T(t), \quad (14)$$

where $x_{\{LJ\}}(t) = 1$ or 0 indicates that whether JAs choose to ally with LUs or not in time slot t , and ξ_n is the unit transmission power cost of each single LU $n \in \mathcal{N}$.

Note that the reason why $()^+$ is used in equation (13) but not in equation (7) is that, in equation (13), the term $W[\cdot]^+$ represents the secrecy transmission rate of LUs without JAs' help, and this rate must be non-negative, however, in equation (7), $V_{\mathcal{K}}^0(t)$ represents the benefit of EVs from intercepting LUs' confidential signals. Since EVs can obtain a positive benefit by successfully intercepting the LUs' confidential signals, the term $W(\log(1 + (g_{nm}(t)P_n^T(t))/n_m^2) - \max_{k \in \mathcal{K}} \log(1 + (g_{nk}(t)P_n^T(t))/n_k^2))$ does not need to be non-negative. Similar formulation has been widely employed in the literature [26], [27], [28].

Denoting LUs' strategies as $\pi_{\mathcal{N}} = \{P_n^T(t), x_{nm}^T(t), \mu_L(t)\}_{\forall t}$, LUs' long-term optimization problem can be formulated as

$$[\mathcal{LP}] : \arg \max_{\pi_{\mathcal{N}}} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t \in [0, T]} U_{\mathcal{N}}(t) \quad (15)$$

$$s.t., R_{nm}^T(t) \geq R_{min}^T, \forall n \in \mathcal{N}, \forall m \in \mathcal{M} \quad (16)$$

$$x_{nm}^T(t) \in \{0, 1\}, \forall n \in \mathcal{N}, \forall m \in \mathcal{M} \quad (17)$$

$$\sum_{m \in \mathcal{M}} x_{nm}^T(t) = 1, \forall n \in \mathcal{N}, \quad (18)$$

$$0 \leq \mu_L(t) \leq \mu_L^{max}, \quad (19)$$

$$P_n^T(t) \in \{P_{n,0}^T, P_{n,1}^T, \dots, P_{n,l-1}^T\}, \forall n \in \mathcal{N}, \quad (20)$$

where constraint (16) states that the transmission rate of each LU in any time slot t should not be smaller than a minimum value for guaranteeing its quality of service (QoS); constraints (17), (18) refer to the many-to-one association between LUs and BSs in any time slot t ; and constraint (20) stands for the set of l -leveled transmission powers of LUs.

For JAs, given $\mu_E(t)$ and $\mu_L(t)$, in each time slot t , they have to determine i) the jamming power allocation $P_j^J(t), \forall j \in \mathcal{J}$; and ii) the coalition selection $x_{\{EJ\}}(t) \in \{0, 1\}$ and $x_{\{LJ\}}(t) \in \{0, 1\}$ to maximize their total reward while reducing their total jamming power consumption, where $x_{\{EJ\}}(t) = 1$ or 0 and $x_{\{LJ\}}(t) = 1$ or 0 mean that JAs choose to form the coalition with EVs or not and with LUs or not, respectively. Intuitively, JAs' reward stems from the incentives obtained from EVs or LUs in each time slot t , denoted as

$$V_{\mathcal{J}}(t) = x_{\{EJ\}}(t)\mu_E(t)(V_{\mathcal{K}}(t) - V_{\mathcal{K}}^0(t)) + x_{\{LJ\}}(t)\mu_L(t)(V_{\mathcal{N}}(t) - V_{\mathcal{N}}^0(t)). \quad (21)$$

Then, JAs' utility in each time slot t can be expressed as

$$U_{\mathcal{J}}(t) = V_{\mathcal{J}}(t) - \eta_j \sum_{j \in \mathcal{J}} P_j^J(t) - c_{conf} \mathcal{F}(t), \quad (22)$$

where η_j is the unit jamming power cost of each single JA $j \in \mathcal{J}$, c_{conf} denotes the potential configuration cost

caused by the additional connection established by JAs to notify the coalition changes if JAs choose to change allies over two consecutive time slots,³ and function $\mathcal{F}(t) = (x_{\{EJ\}}(t), x_{\{LJ\}}(t)) \oplus (x_{\{EJ\}}(t-1), x_{\{LJ\}}(t-1))$ indicates whether the coalitions in time slot t change or not compared to those in previous time slot $t-1$.

With JAs' strategy denoted as $\pi_{\mathcal{J}} = \{P_j^J(t), x_{\{EJ\}}(t), x_{\{LJ\}}(t)\}_{\forall t}$, the long-term optimization problem for JAs can be formulated as

$$[\mathcal{JP}] : \arg \max_{\pi_{\mathcal{J}}} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t \in [0, T]} U_{\mathcal{J}}(t) \quad (23)$$

$$s.t., x_{\{EJ\}}(t), x_{\{LJ\}}(t) \in \{0, 1\}, \quad (24)$$

$$x_{\{EJ\}}(t) + x_{\{LJ\}}(t) \leq 1 \quad (25)$$

$$P_j^J(t) \in \{P_{j,0}^J, P_{j,1}^J, \dots, P_{j,l-1}^J\}, \forall j \in \mathcal{J} \quad (26)$$

where constraints (24), (25) signify that JAs can only cooperate unilaterally with either LUs or EVs in each time slot; and constraint (26) represents the l -leveled jamming powers of JAs.

Obviously, problems $[\mathcal{EP}]$, $[\mathcal{LP}]$ and $[\mathcal{JP}]$ are tightly coupled. In each time slot, we can allow EVs to first determine $x_k^{sw}(t)$ and $\mu_E(t)$, and then as the response, LUs can determine $P_n^T(t)$, $x_{nm}^T(t)$ and $\mu_L(t)$, and finally given the strategies of EVs and LUs, JAs in turn can determine $P_j^J(t)$, $x_{\{LJ\}}(t)$ and $x_{\{EJ\}}(t)$. The decision-making sequence of these three parties in each time slot is inspired from the fact that LUs tend to respond to EVs' decisions for increasing their secrecy transmission rate, and JAs tend to make decisions of choosing the side to assist in exchange for a higher reward. Similar decision-making structure has been widely investigated in [26] and [29]. Formally, the strategic interactions among EVs, LUs and JAs in PLS can be modeled by a three-stage hierarchical game $\mathcal{G}^H$, defined as

$$\mathcal{G}^H = \{\mathcal{K}, \mathcal{N}, \mathcal{J}, \pi_{\mathcal{K}}, \pi_{\mathcal{N}}, \pi_{\mathcal{J}}, U_{\mathcal{K}}, U_{\mathcal{N}}, U_{\mathcal{J}}\}, \quad (27)$$

where $\mathcal{K}, \mathcal{N}, \mathcal{J}$ refer to EVs, LUs and JAs, respectively, $\pi_{\mathcal{K}}, \pi_{\mathcal{N}}, \pi_{\mathcal{J}}$ denote the aforementioned strategies of these three parties, respectively, and $U_{\mathcal{K}}, U_{\mathcal{N}}, U_{\mathcal{J}}$ represent the utility functions of these three parties, respectively. Note that our proposed dynamic hierarchical game $\mathcal{G}^H$ not only repeats over different time slots, but also iterates the decision sequence of LUs, EVs and JAs within one time slot until the convergence. Although we state that LUs make decision after EVs and JAs are the last to make decisions, this is only an initial sequence in a single round, and has to iterate multiple rounds until the convergence. Due to this back-and-forth iterations, there is actually no unique sequence for the decisions of the three parties, and making decisions following any sequence will have the same performance.

Fig. 2 illustrates such repeated decision-making process of $\mathcal{G}^H$ and the leader-follower relationships among them in each time slot, of which the stage 1, stage 2 and stage 3 refer to the decision-making rounds of EVs, LUs and JAs, respectively.

³Note that, in this paper, coalitions are only required to be stable within a certain time slot, and can vary dynamically over different time slots, while an additional configuration cost has to be introduced whenever a coalition change occurs. This further implies that the coalition states among multiple time slots are interdependent.

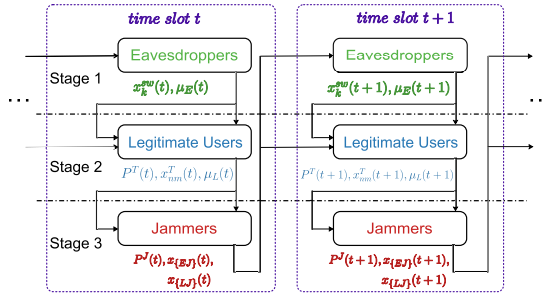


Fig. 2. Repeated decision-making process of the proposed $\mathcal{G}^H$.

Moreover, it can be observed that the formulated hierarchical game $\mathcal{G}^H$ actually involves dynamic trilateral coalitions among three parties (i.e., EVs, LUs and JAs) in PLS, i.e., LUs and JAs may form the coalition to against EVs in one time slot, while EVs and JAs may form the coalition to against LUs in another, making it very difficult to be solved directly. Specifically, the coalition selection of these three parties are mutually and dynamically decided by each party, and the study on how each party decide the coalition it belongs to and how such coalition selection changes over different time slots with the strong interdependence are of great significance. In the following section, by first introducing the underlying trilateral coalition formation game, we analyze the stability and equilibrium conditions, and then propose a DRL-based solution for deriving the equilibrium of $\mathcal{G}^H$ in dynamic evolutions with long-term performance guarantees.

IV. GAME ANALYSIS AND DRL-BASED SOLUTION

In this section, we first define the underlying CFG among all three parties (i.e., EVs, LUs and JAs) in PLS, along with the conditions for a single party to join a certain coalition. Then, $\mathcal{G}^H$ formulated in Section III is re-defined and analyzed. Afterwards, we propose an HCSF algorithm for deriving the stable coalition partition in each time slot, while ensuring the existence of $\mathcal{G}^H$'s equilibrium in the repeated decision-making process. Finally, a DRL-based solution is developed to produce the optimal strategies for all three parties in every time slot with long-term performance guarantees.

A. Trilateral Coalition Formation Game

To better describe the dynamic coalition formation among LUs, EVs and JAs in PLS, we particularly design a trilateral CFG, defined as

$$\mathcal{G}^C = \{G, \Delta, U_K, U_N, U_J\}, \quad (28)$$

where $G = \{K, N, J\}$ denotes the set of players, i.e., three parties (EVs, LUs and JAs) in PLS, U_K, U_N, U_J respectively denote their utility functions, as described in the previous section, and $\Delta = \{\{K\}, \{N\}, \{J\}, \{K, J\}, \{N, J\}\}$ represents all possible coalitions among them.

Each party has different preferences for joining different potential coalitions. Specifically, the condition under which party $i \in G$ prefers to join a possible coalition $C_a \in \Delta \cup \{\emptyset\}$ rather than another one $C_b \in \Delta \cup \{\emptyset\}$ in time slot t can be

expressed as

$$C_a \succ_i^t C_b \text{ if and only if } U_i^{C_a \cup i}(t) > U_i^{C_b \cup i}(t), \\ \forall C_a, C_b \in \Delta \cup \{\emptyset\}, C_a \neq C_b, \quad (29)$$

where symbol $\succ_i^t$ denotes the coalition preference order [11], [30] for each party $i \in G$ in time slot t , and $U_i^{C_a \cup i}(t)$ and $U_i^{C_b \cup i}(t)$ represent party i 's utilities after joining coalition C_a and C_b , respectively.

Although the above preference order may guide each party (i.e., LUs, EVs and JAs) to choose the most beneficial coalition to join, since this order has to be applied to all three parties, any coalition partitions formed without considering all allies' selfishness within the coalition is unstable. For example, if J selfishly choose to form a coalition with LUs, i.e., join $\{N\}$ in time slot t in case that $\{N\} \succ_J^t \{K\}$, but LUs' utility decreases due to JAs' joining, such coalition partition $\{\{J, N\}, \{K\}\}$ does not hold because the preference order for LUs becomes $\emptyset \succ_N^t \{J\}$. Thus, we call a coalition partition is stable if and only if no subset of players in G would secede from its original coalition and form any other coalitions (with higher utility gains). Accordingly, the coalition switch rule [17] for describing the condition of each single party's leaving its original coalition $C_a \in \Delta$ and joining in another new coalition $C_b \in \Delta$ in time slot t can be expressed as

$$C_a \rightarrow_i^t C_b \text{ if and only if } C_b \succ_i^t C_a \setminus i \text{ \& } C_b \cup i \setminus i' \succ_i^t C_b \setminus i', \\ \forall i \in C_a, \forall i' \in C_b, \forall C_a, C_b \in \Delta, C_a \neq C_b, \quad (30)$$

where $\rightarrow_i^t$ indicates party i 's coalition switch in time slot t . Besides, as a special case, if party i 's utility after joining any coalition is smaller than that of being alone, then it will choose to not ally with any other parties, which can be written as

$$C_a \rightarrow_i^t \emptyset \text{ if and only if } \emptyset \succ_i^t C_a \setminus i, \forall i \in C_a, \forall C_a \in \Delta. \quad (31)$$

Based on these, we are now able to re-define the hierarchical game $\mathcal{G}^H$ formulated in (27) as follows.

Definition 1 ($\mathcal{G}^H$ Integrated With $\mathcal{G}^C$): With the underlying CFG $\mathcal{G}^C = \{G, \Delta, U_K, U_N, U_J\}$ defined in (28), game $\mathcal{G}^H$ can be reformulated as $\mathcal{G}^H = \{G, \Pi^H, \Pi^C, \mathcal{U}\}$, where $\Pi^H = \{\pi_i^H\}_{\forall i \in G}$, in which $\pi_K^H = \{x_k^{sw}(t), \mu_E(t)\}_{\forall t}$, $\pi_N^H = \{x_{nm}^T(t), P_n^T(t), \mu_L(t)\}_{\forall t}$ and $\pi_J^H = \{P_j^J(t)\}_{\forall t}$; $\Pi^C = \{\pi_i^C\}_{\forall i \in G}$, in which $\pi_i^C = \{\delta_i(t)\}_{\forall t}$, and $\delta_i(t) \in \Delta$ signifies the coalition selection decision of each party $i \in G$ in time slot t ; and $\mathcal{U} = \{U_K, U_N, U_J\}$.

B. Analysis on Hierarchical Game

In this subsection, we analyze the equilibrium of the formulated game $\mathcal{G}^H$ and show its existence. For each party $i \in G$, let (π_i^{H*}, π_i^{C*}) be its best response in hierarchical game $\mathcal{G}^H$, indicating its optimal strategies, which are also equivalent to the solution of $[\mathcal{EP}]$, $[\mathcal{LP}]$ and $[\mathcal{JP}]$. Then, the equilibrium of game $\mathcal{G}^H$ can be defined as follows.

Definition 2 (Equilibrium of $\mathcal{G}^H$): In the proposed game $\mathcal{G}^H$, a strategy profile (π_i^{H*}, π_i^{C*}) , $\forall i \in G$, is the equilibrium if for every party $i \in G$, we have

$$\frac{1}{T} \sum_{t=0}^{T-1} U_i(t) | \pi_i^{H*}, \pi_i^{C*}, \pi_{-i}^{H*}, \pi_{-i}^{C*} \\ \geq \frac{1}{T} \sum_{t=0}^{T-1} U_i(t) | \pi_i^H, \pi_i^C, \pi_{-i}^{H*}, \pi_{-i}^{C*}, \quad (32)$$

where π_{-i}^{H*} and π_{-i}^{C*} denote the equilibrium strategies of all other parties except i . Obviously, when such equilibrium is reached, the long-term utility of each party can be maximized, and no party will deviate this equilibrium unilaterally.

Observe that in the above definition, the equilibrium of game $\mathcal{G}^H$ highly depends on π_i^{C*} , $\forall i \in G$, which is the equilibrium of the underlying CFG $\mathcal{G}^C$. Therefore, we are required to explore the equilibrium condition of game $\mathcal{G}^C$, also known as its stability condition. In time slot t , let $\mathcal{CP}(t) = \bigcup_{i \in G} \{\delta_i(t)\}$ represent the coalition partition that LUs, EVs and JAs choose to form. Then, the equilibrium of $\mathcal{G}^C$, i.e., the stable coalition partition, is defined as follows.

Definition 3 (Stable coalition partition of $\mathcal{G}^C$): In each time slot t , a coalition partition $\mathcal{CP}^*(t) = \bigcup_{i \in G} \{\delta_i^*(t)\}$ is stable if no party can improve its utility by changing its coalition unilaterally (i.e., leaving its original coalition and joining the others), meaning that

$$U_i^{C_a}(t) \geq U_i^{C_b \cup i}(t), \forall C_a, C_b \in \mathcal{CP}^*(t), C_a \neq C_b, \forall i \in C_a.$$

It is worth noting that this definition mainly focuses on describing the stability requirement for a coalition partition in each time slot, while does not prohibit the dynamic changes of coalition partitions over different time slots. Next, we show that the proposed CFG $\mathcal{G}^C$ is an exact potential game (EPG) [15], and thus there exists at least one pure Nash equilibrium or equivalently a stable coalition partition.

Theorem 1 (Existence of the Stable Coalition Partition): The proposed CFG $\mathcal{G}^C$ exists a stable coalition partition in each time slot t .

Proof: Similar to those in [15] and [17], for game $\mathcal{G}^C$, we define the system utility of each party i of joining coalition C_a as

$$u_i^t(C_a, C_{-a}) = U^t(C_a) - U^t(C_a \setminus i), \forall i \in G, \quad (33)$$

where $U^t(C_a) = \sum_{i \in C_a} U_i(t)$ stands for the total utility of all members in C_a and $U^t(C_a \setminus i)$ denotes the total utility of all members in C_a except i , so that $u_i^t(C_a, C_{-a})$ calculates the contribution of party i to the system utility by choosing to join coalition C_a . Then, we can design a corresponding potential function as

$$\zeta^t(C_a, C_{-a}) = \sum_{i \in C_a} u_i^t(C_a, C_{-a}), i \in C_a. \quad (34)$$

Suppose that a party $i \in G$ chooses to switch from coalition C_a to $C_{a'}$, the difference brought to the potential function can be derived as

$$\begin{aligned} \zeta^t(C_a, C_{-a}) - \zeta^t(C_{a'}, C_{-a'}) &= U^t(C_a) + U^t(C_{a'} \setminus i) + U^t(C_{others}) - U^t(C_{a'}) \\ &\quad - U^t(C_a \setminus i) - U^t(C_{others}) \\ &= u_i(C_a, C_{-a}) - u_i(C_{a'}, C_{-a'}), \end{aligned} \quad (35)$$

where C_{others} stands for the other coalitions formed by parties except those in C_a or $C_{a'}$. Obviously, the difference on potential functions equals the difference on system utilities owing to i 's coalition switching. Therefore, following the literature [15], [17], the proposed CFG $\mathcal{G}^C$ is an EPG. Moreover, since it is widely proved that EPG has at least one pure Nash equilibrium, we can conclude that the equilibrium of $\mathcal{G}^C$ in

Algorithm 1 Hedonic Coalition Selection and Formation Algorithm (HCSF)

Input: Observed EVs' decision $\mu_E(t)$, observed LUs' decisions $P_n^T(t)$ and $\mu_L(t)$, channel gains $\{g_{ij}(t)\}_{\forall i, \forall j}$, coalition partition $x_{\{EJ\}}(t-1)$ and $x_{\{LJ\}}(t-1)$.

Output: Final coalition partition $\mathcal{CP}(t)$, $x_{\{EJ\}}(t)$ and $x_{\{LJ\}}(t)$.

- 1 Initialize: $x_{\{EJ\}}(t) = x_{\{EJ\}}(t-1)$ and $x_{\{LJ\}}(t) = x_{\{LJ\}}(t-1)$;
- 2 **while** $\mathcal{CP}(t)$ is different from it in the previous loop **do**
- 3 **for** party $i \in \{\mathcal{K}, \mathcal{N}, \mathcal{J}\}$ **do**
- 4 **for** $C_{ob}, i \notin C_{ob}, C_{ob} \in \mathcal{CP}(t) \cup \{\emptyset\}$ **do**
- 5 Player i belongs to coalition C_{oa} , calculate $U_i^{C_{oa}}(t), U_i^{C_{ob}}(t)$ according to (8), (14) or (22);
- 6 Player i choose to i) leaves its current coalition and join a new coalition; or ii) stays in its current coalition according to single player's switch rules (30), (31) and its preference order (29);
- 7 Adjust the coalition partition $\mathcal{CP}(t)$;
- 8 Update the $x_{\{EJ\}}(t)$ and $x_{\{LJ\}}(t)$ according to $\mathcal{CP}(t)$.

each time slot exists (i.e., $\mathcal{G}^C$'s stable coalition partition in each time slot exists). ■

Further inspired by [11] and [17], we propose an HCSF algorithm based on the introduced switch rules (30), (31) to derive such stable coalition partition in each time slot. Specifically, given the coalition partition inherited from the previous time slot, i.e., $\mathcal{CP}(t-1)$, in each existing coalition $C_a \in \mathcal{CP}(t-1)$, each party $i \in C_a$ first calculates its utility, and then decides to potentially switch its coalition according to switch rules (30), (31). This process repeats for all parties in every coalition until the coalition partition is stabilized. We denote such stable coalition partition as $\mathcal{CP}(t)$ and its corresponding solution as $x_{\{LJ\}}(t)$ and $x_{\{EJ\}}(t)$. Algorithm 1 summarizes the detailed procedure of the proposed HCSF algorithm.

It is worth noting that, different from the conventional coalition partition or formation approaches in [11], [16], and [31], following the definition in [32], the proposed coalition selection and formation algorithm HCSF is hedonic, namely, i) the stable coalition partition in each time slot is formed based on each party's preference order and the coalition switch rules defined in (29)-(31); and ii) the utility of each party depends on not only itself but also its allies. Therefore, a low computational complexity can be achieved by the proposed HCSF algorithm. To be more specific, the computational complexity of the proposed HCSF algorithm is $O(|C_{oi}| * |\Delta|)$, which is lower than those of the centralized merge-and-split algorithm [11] and epsilon core based algorithm [31], with computational complexities being $O(|G|^3)$ and $O(|G|^{|\Delta|})$, respectively.

Theorem 2 (Convergence of HCSF Algorithm): By adopting the proposed HCSF algorithm summarized in Algorithm 1, all three parties in PLS (i.e., $\mathcal{K}, \mathcal{N}, \mathcal{J}$) will eventually converge to a hedonic coalition partition $\mathcal{CP}^*(t)$ in each time slot t .

Proof: Defining the utility of any coalition $C_a \in \mathcal{CP}(t)$ as $v(C_a) = \sum_{i \in C_a} U_i^{C_a}(t)$, it is shown in literature [33] that a coalition partition $\mathcal{CP}(t)$ is stable if and only if the following two conditions are satisfied:

- 1) the utility of coalition C_a in $\mathcal{CP}(t)$ is larger than or equal to the total utility of all sub-coalitions in any possible partition of C_a , denoted as

$$v(C_a) \geq \sum_b v(C'_b), \forall C_a \in \mathcal{CP}(t), \bigcup_b C'_b = C_a; \quad (36)$$

- 2) the total utility of all coalitions $\mathcal{CP}(t)$ is larger than or equal to the utility of the grand coalition of all parties, expressed as

$$\sum_a v(C_a) \geq v(\bigcup_{i \in G} \{i\}). \quad (37)$$

For $\mathcal{CP}^*(t)$ obtained by the proposed HCSF algorithm, there are three possible cases, and in each case, we can prove that the above stability conditions hold.

Case 1: All three parties choose to not form any coalition with the others in time slot t , i.e., $\mathcal{CP}^*(t) = \{\mathcal{J}, \mathcal{K}, \mathcal{N}\}$. In this case, all elements in $\mathcal{CP}^*(t)$ are singletons, and thus inequality (36) must be satisfied;

Case 2: JAs and LUs choose to form a coalition in time slot t , i.e., $\mathcal{CP}(t) = \{(\mathcal{J}, \mathcal{N}), \mathcal{K}\}$. In this case, according to switch rules (30) and (31), there are two subcases described as follows.

- **Subcase 2.1:** JAs originally cooperated with EVs, but now switch to aid LUs at time slot t , i.e., $(\mathcal{J}, \mathcal{K}) \rightarrow_{\mathcal{J}}^t \mathcal{N}$. In this subcase, $\mathcal{J} \succ_{\mathcal{N}}^t \emptyset$ and $\mathcal{N} \succ_{\mathcal{K}}^t \mathcal{K}$ both hold according to (30). Then, we have $U_{\mathcal{N}}^{\mathcal{J}, \mathcal{N}}(t) > U_{\mathcal{N}}^{\mathcal{K}}(t)$ and $U_{\mathcal{J}}^{\mathcal{J}, \mathcal{N}}(t) > U_{\mathcal{J}}^{\mathcal{K}}(t)$ according to (29). Since the coalition $(\mathcal{J}, \mathcal{K})$ exists in the coalition formation process, inequality $\mathcal{K} \succ_{\mathcal{J}}^t \emptyset$ is satisfied, which means $U_{\mathcal{J}}^{\mathcal{J}, \mathcal{K}}(t) > U_{\mathcal{J}}^{\mathcal{J}}(t)$. Therefore, we have $U_{\mathcal{N}}^{\mathcal{J}, \mathcal{N}}(t) + U_{\mathcal{J}}^{\mathcal{J}, \mathcal{N}}(t) > U_{\mathcal{J}}^{\mathcal{J}, \mathcal{K}}(t) + U_{\mathcal{N}}^{\mathcal{K}}(t) > U_{\mathcal{J}}^{\mathcal{J}}(t) + U_{\mathcal{N}}^{\mathcal{N}}(t)$, which is equivalent to $v(\mathcal{J}, \mathcal{N}) > v(\mathcal{J}) + v(\mathcal{N})$, and hence inequality (36) must be satisfied.
- **Subcase 2.2:** JAs and LUs were both non-cooperative, but now mutually switch to form a coalition at time slot t , i.e., $\mathcal{J} \rightarrow_{\mathcal{J}}^t \mathcal{N}$ and $\mathcal{N} \rightarrow_{\mathcal{N}}^t \mathcal{J}$ both hold. In this subcase, $\mathcal{N} \succ_{\mathcal{J}}^t \emptyset$ and $\mathcal{J} \succ_{\mathcal{N}}^t \emptyset$ both hold according to (30). Then, we have $U_{\mathcal{J}}^{\mathcal{J}, \mathcal{N}}(t) > U_{\mathcal{J}}^{\mathcal{J}}(t)$ and $U_{\mathcal{N}}^{\mathcal{J}, \mathcal{N}}(t) > U_{\mathcal{N}}^{\mathcal{N}}(t)$ according to (29). Therefore, we have $U_{\mathcal{N}}^{\mathcal{J}, \mathcal{N}}(t) + U_{\mathcal{J}}^{\mathcal{J}, \mathcal{N}}(t) > U_{\mathcal{J}}^{\mathcal{J}}(t) + U_{\mathcal{N}}^{\mathcal{N}}(t)$, which is equivalent to $v(\mathcal{J}, \mathcal{N}) > v(\mathcal{J}) + v(\mathcal{N})$, and hence inequality (36) must be satisfied.

Case 3: JAs and EVs choose to form a coalition in time slot t , i.e., $\mathcal{CP}(t) = \{(\mathcal{J}, \mathcal{K}), \mathcal{N}\}$. In this case, we can apply the similar argument as in the proof for Case 2 to show that inequality (36) holds. The only difference is that $\mathcal{K}$ is replaced with $\mathcal{N}$.

Note that the grand coalition, i.e., $\{(\mathcal{J}, \mathcal{N}, \mathcal{K})\}$ does not exist in the formulated problem because LUs and EVs inherently conflict with each other, and thus condition (37) always holds in all three cases.

To sum up, $\mathcal{CP}^*(t)$ formed by the proposed HCSF algorithm satisfies both (36) and (37). This completes the proof. ■

Since a pure-strategy equilibrium for hierarchical game $\mathcal{G}^H$ may not always exist or is difficult if not impossible to be obtained [17], we consider a more general solution by defining the mixed strategies of EVs, LUs and JAs in game $\mathcal{G}^H$ as $Z_{\mathcal{K}} = \Gamma(\{x_k^{sw}(t), \mu_E(t)\})$, $Z_{\mathcal{N}} = \Theta(\{x_{nm}^T(t), P_n^T(t), \mu_L(t)\})$ and $Z_{\mathcal{J}} = \Lambda(\{P_j^J(t)\})$, where $\Gamma(\cdot)$, $\Theta(\cdot)$ and $\Lambda(\cdot)$ denote the probability distribution over EVs', LUs, and JAs' deterministic strategies, respectively. Correspondingly, the equilibrium of $\mathcal{G}^H$ defined in Definition 2 (i.e., $\pi_{\mathcal{K}}^{H*}$, $\pi_{\mathcal{N}}^{H*}$ and $\pi_{\mathcal{J}}^{H*}$) can be converted to a probability form (i.e., $Z_{\mathcal{K}}^*$, $Z_{\mathcal{N}}^*$ and $Z_{\mathcal{J}}^*$).

Theorem 3 (Existence of the Mixed Strategy Equilibrium for $\mathcal{G}^H$): In the proposed hierarchical game $\mathcal{G}^H$, there exists at least one mixed strategy equilibrium (Z_i^*, π_i^{C*}) , where $Z_i^*, \forall i \in G$ is the mixed strategy form of $\{\pi_i\}_{\forall i}$, and π_i^{C*} is the equilibrium strategy of dynamic coalition formation in all time slots of $\mathcal{G}^H$. Mathematically, such mixed strategy equilibrium (Z_i^*, π_i^{C*}) satisfies

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} U_i(t) | Z_i^*, Z_{-i}^*, \pi_i^{C*}, \pi_{-i}^{C*} \\ \geq \frac{1}{T} \sum_{t=0}^{T-1} U_i(t) | Z_i, Z_{-i}, \pi_i^C, \pi_{-i}^{C*}, \end{aligned} \quad (38)$$

where Z_{-i}^* denotes the equilibrium mixed strategies of all other parties except i .

Proof: First, we prove that, given the strategy $\{\pi_i^H\}_{\forall i \in G}$, the stable coalition partition $\mathcal{CP}^*(t)$ generated by the proposed HCSF algorithm is also the subgame perfect equilibrium (SPE) of $\mathcal{G}^H$. Denoting $\{\pi_{i,t}^{C*}\}_{\forall i \in G}$ as all three parties' local equilibrium strategy which is equivalent to $\mathcal{CP}^*(t)$, then according to Definition 3, given the strategy (π_i^H, π_{-i}^H) for hierarchical game $\mathcal{G}^H$ without CFG $\mathcal{G}^C$, we have

$$\begin{aligned} U_i(t) | \pi_i^H, \pi_{-i}^H, \pi_{i,t}^{C*}, \pi_{-i,t}^{C*} \geq U_i(t) | \pi_i^H, \pi_{-i}^H, \pi_{i,t}^C, \pi_{-i,t}^{C*}, \\ \forall i \in G, \forall t \in [0, 1, \dots, T-1], \end{aligned} \quad (39)$$

where $\pi_{-i,t}^{C*}$ is the local equilibrium strategy for $\mathcal{G}^C$ of other parties except i in time slot t , and $\pi_{i,t}^C$ is a non-equilibrium strategy for $\mathcal{G}^C$ of party i in time slot t .

Denoting the subgame of $\mathcal{G}^H$ from time slot $\tau \in \{0, 1, \dots, T-1\}$ to $T-1$ as $\mathcal{G}^H[\tau]$. By using the backward induction [34], in any subgame $\mathcal{G}^H[\tau]$, $\forall \tau \in \{0, 1, \dots, T-1\}$, the following inequality must hold:

$$\begin{aligned} \frac{1}{T-\tau} \sum_{t=\tau}^{T-1} U_i(t) | \pi_i^H, \pi_{-i}^H, \pi_{i,t}^{C*}, \pi_{-i,t}^{C*} \\ \geq \frac{1}{T-\tau} \sum_{t=\tau}^{T-1} U_i(t) | \pi_i^H, \pi_{-i}^H, \pi_{i,t}^C, \pi_{-i,t}^{C*}, \forall i \in G. \end{aligned} \quad (40)$$

The above inequality indicates that, given the strategy (π_i^H, π_{-i}^H) , in any subgame $\mathcal{G}^H[\tau]$, $\forall \tau \in \{0, 1, \dots, T-1\}$, the average utility of each party $i \in G$ achieved by $\mathcal{G}^H[\tau]$'s coalition formation strategy $\{\pi_{i,t}^{C*}\}_{t=\tau}^{T-1}$ obtained by the proposed HCSF algorithm is higher than that achieved by any other coalition formation strategy $\{\pi_{i,t}^C\}_{t=\tau}^{T-1}$, and thereby

$\{\pi_{i,t}^{C*}, \pi_{-i,t}^{C*}\}_{t=\tau}^{T-1}$ is the equilibrium strategy of $\mathcal{G}^H[\tau]$. In other words, given the strategy (π_i^H, π_{-i}^H) , the strategy profile $\{\pi_{i,t}^{C*}, \pi_{-i,t}^{C*}\}_{\forall t}$ constitutes the SPE of the overall $\mathcal{G}^H$ in all time slots [34], [35]. Therefore, $\{\pi_{i,t}^{C*}\}_{\forall t}$ becomes equivalent to π_i^{C*} as defined in Definition 2. By replacing $\{\pi_{i,t}^{C*}\}_{\forall t}$ with π_i^{C*} , since π_i^{C*} is the SPE of $\mathcal{G}^H$ given the strategy $\{\pi_i^H\}_{\forall i \in G}$, the following inequality must hold:

$$\begin{aligned} & \frac{1}{T} \sum_{t=0}^{T-1} U_i(t) | \pi_i^H, \pi_{-i}^H, \pi_i^{C*}, \pi_{-i}^{C*} \\ & \geq \frac{1}{T} \sum_{t=0}^{T-1} U_i(t) | \pi_i^H, \pi_{-i}^H, \pi_i^C, \pi_{-i}^{C*}, \forall i \in G. \end{aligned} \quad (41)$$

From inequality (41), we can observe that the existence of hierarchical game $\mathcal{G}^H$'s equilibrium also depends on the existence of equilibrium strategy of (π_i^H, π_{-i}^H) . Hence, the remaining problem is to prove that the equilibrium solution exists in the proposed hierarchical game $\mathcal{G}^H$ without $\mathcal{G}^C$. This is intuitive because i) the number of all parties is finite; ii) the strategies of all three parties (i.e., all decision variables in $\{\pi_i\}_{\forall i \in G}$ are discrete and limited) are finite; and iii) the number of time slots (i.e., $t \in [0, 1, \dots, T]$) can be set as finite, meaning that the proposed hierarchical game $\mathcal{G}^H$ without $\mathcal{G}^C$ is a finite game which must have a mixed strategy equilibrium according to [17] and [36]. Denoting the mixed strategy equilibrium of $\mathcal{G}^H$ without $\mathcal{G}^C$ as (Z_i^*, Z_{-i}^*) , where Z_i^* is the mixed strategy of party, and Z_{-i}^* is the mixed strategy of other parties except i .

With the equilibrium strategy $(\pi_i^{C*}, \pi_{-i}^{C*})$ achieved by the proposed HCSF algorithm, inequality (41) can be further derived as

$$\begin{aligned} & \frac{1}{T} \sum_{t=0}^{T-1} U_i(t) | Z_i^*, Z_{-i}^*, \pi_i^{C*}, \pi_{-i}^{C*} \\ & \geq \frac{1}{T} \sum_{t=0}^{T-1} U_i(t) | Z_i, Z_{-i}^*, \pi_i^C, \pi_{-i}^{C*}, \forall i \in G. \end{aligned} \quad (42)$$

This completes the proof. ■

C. DRL-Based Solution for Hierarchical Game

Owing to the observations that the strategies of each party, i.e., LUs, EVs and JAs in PLS in each time slot only depend on those strategies and resulted system states, e.g., the coalition states and channel states in the previous time slot (meaning that state transitions satisfy the Markov property), we can characterize the strategy making problems for LUs, EVs and JAs by three respective Markov decision processes (MDPs). On top of this, we propose a DRL-based solution for sequentially training all three parties to obtain the mixed strategy equilibrium $Z_i^*, \forall i \in G$ of game $\mathcal{G}^H$, as defined in Theorem 3.

MDP for Each Party in PLS: For each party $i \in G = \{\mathcal{K}, \mathcal{N}, \mathcal{J}\}$ in PLS, its corresponding MDP is expressed as $\mathcal{M}_i = \{\mathcal{S}_i, \mathcal{A}_i, \Xi_i, \mathcal{R}_i\}$.

1) Environment State for Each Party in PLS: For each party $i \in G$ in time slot t , its environment state can be expressed as $s_i^t = \{\mathcal{CP}(t), \mathcal{G}(t), A_{-i}(t)\}$, where $\mathcal{CP}(t)$ is i 's current available coalition partition, $\mathcal{G}(t) = \{g_{xy}(t)\}_{\forall x \in \mathcal{N} \cup \mathcal{J}, \forall y \in \mathcal{M} \cup \mathcal{K}}$ denotes channel gains of all possible links, and $A_{-i}(t) = \{a_{-i}\}_{\forall a_{-i} \in \mathcal{A}_{-i}}$ indicates all other parties' actions except i (which will be defined later). Let $\mathcal{S}_i = \{s_i^t\}_{\forall t}$ stand for

Algorithm 2 DRL-Based Solution for Game $\mathcal{G}^H$

Input: Initial locations of all devices.
Output: The equilibrium, i.e., $Z_{\mathcal{K}}^*, Z_{\mathcal{N}}^*, Z_{\mathcal{J}}^*$.

- 1 Initialize: $NT(0)$, actor and critic networks of all agents, and $\mathcal{CP}(0) = \{x_{\{EJ\}}(0), x_{\{LJ\}}(0)\} = \{0, 0\}$;
- 2 **for** training step = 0, 1, 2, ..., max **do**
- 3 **for** $t = 1, 2, \dots, T$ **do**
- 4 $\mathcal{CP}(t) = \mathcal{CP}^*(t-1)$;
- 5 $s_0 = s = \{NT(t), \mathcal{CP}(t)\}$;
- 6 $agt_{sw}^{\mathcal{K}}$ generates action $x_k^{sw}(t)$ based on s ;
- 7 Update $NT(t)$ by $x_k^{sw}(t)$;
- 8 $agt_{\mu}^{\mathcal{K}}$ generates action $\mu_E(t)$ based on s ;
- 9 **for** $n = 1, 2, \dots, N$ **do**
- 10 $agt_n^{\mathcal{N}}$ generates $P_n^T(t)$ and $x_{nm}^T(t)$ based on s ;
- 11 Update $NT(t)$ by $P_n^T(t)$ and $x_{nm}^T(t)$;
- 12 $agt_{\mu}^{\mathcal{N}}$ generates action $\mu_L(t)$ based on s ;
- 13 $agt_P^{\mathcal{J}}$ generates action P_j^J based on $s, \mu_E(t), \mu_L(t)$;
- 14 **for each coalition** $C_a \in \mathcal{CP}(t)$ **do**
- 15 **for each party** $i \in C_a$ **do**
- 16 Call Algorithm 1 to get $\mathcal{CP}^*(t)$;
- 17 Update $\mathcal{CP}(t)$;
- 18 $s = \{NT(t), \mathcal{CP}(t)\}$;
- 19 Calculate each party's rewards $r_i^t, \forall i \in G$;
- 20 **for each agent** $e \in \mathbb{AGT}$ **do**
- 21 Store $s_0, a_e^t, r_e^t = r_i^t$ and s in e 's replay buffer, where i is e 's corresponding party;
- 22 **if replay buffers reach maximum capacity then**
- 23 **for each agent** $e \in \mathbb{AGT}$ **do**
- 24 Calculate e 's rewards-to-go
- 25 $J_e^t = \sum_{t=0}^{T-1} \gamma^t r_e^t$;
- 26 Calculate e 's advantage $A_e^t = J_e^t - V_{\phi_e}^e(s_t^e)$;
- 27 Generate the loss for e 's actor network: $L_{\theta}^e = \sum_{t=0}^{T-1} \min(\frac{p_{\theta_e}(a_e^t | s_t^e)}{p_{\theta_e'}(a_e^t | s_t^e)} A_e^t, \text{clip}(\frac{p_{\theta_e}(a_e^t | s_t^e)}{p_{\theta_e'}(a_e^t | s_t^e)}, 1 - \epsilon, 1 + \epsilon) A_e^t)$;
- 28 Generate the loss for e 's critic network: $L_{\phi_e}^e = \sum_{t=0}^{T-1} (V_{\phi_e}^e(s_t^e) - J_e^t)^2$;
- 29 Update ϵ_e and ϕ_e via Adam Optimizer.

party i 's state space. It is obvious that such state space is multi-dimensional, and thus may lead to a potential curse of dimensionality. To reduce dimensions, we define that $s_i^t = \{NT(t), \mathcal{CP}^*(t-1)\}, \forall i \in \{\mathcal{N}, \mathcal{K}\}$, and $s_{\mathcal{J}}^t = \{NT(t), \mathcal{CP}^*(t-1), \mu_E(t), \mu_L(t)\}$, in which $NT(t)$ is an adjacency matrix defined as follows.

$$NT(t) = \begin{bmatrix} x_{nm}^T(t) P_n^T(t) g_{nm}(t) & x_k^{sw}(t) P_n^T(t) g_{nk}(t) \\ P_j^J(t) g_{jm}(t) & x_k^{sw}(t) P_j^J(t) g_{jk}(t) \end{bmatrix}. \quad (43)$$

It is worth noting that the environmental state represents the state information available to each party, and such information can be acquired because of the practical wireless network

settings. Specifically, i) $x_k^{sw}(t)$ can be obtained by LUs and JAs based on local oscillator power leaked by activated EVs' RF front-end using the detection/location method developed in [37] and [38]; ii) transmission/jamming power of LUs/JAs, i.e., $P_n^T(t)$ and $P_j^J(t)$, can be estimated by EVs, LUs and JAs according to the received signal strength based on compressive sensing [39] or kalman filtering [40]; iii) the global channel gains $g_{nm}(t), g_{nk}(t), g_{jm}(t), g_{jk}(t)$ are observable to all parties by utilizing techniques based on the well-known pilot information [41] or torch method [42]; and iv) the unit incentive $\mu_E(t)$ and $\mu_L(t)$ should be broadcasted by EVs and LUs respectively for attracting JA's alliance, which are available to JAs. Moreover, owing to the inherent adversary relationship (i.e., LUs and EVs are always mutually conflict with each other), the grand coalition $\{(\mathcal{K}, \mathcal{N}, \mathcal{J})\}$ never exists. Therefore, each party should and will be willing to broadcast whether it is a singleton to attract other parties' cooperation at the beginning of each time slot. Based on the broadcasted singleton status of each party at time slot t and the prior knowledge of their possible coalition formation at time slot $t-1$ with other parties, each party can calculate $\mathcal{CP}^*(t-1)$ at time slot t .

2) *Action for Each Party in PLS*: In time slot t , each party i 's action is exactly its strategic decisions.⁴ Hence, we have the action space $\mathcal{A}_i = \{\pi_i^H\}_{\forall i \in G}$.

3) *State Transition Probabilities of Each Party in PLS*: The state transition probability from state $s \in \mathcal{S}_i$ to state $s' \in \mathcal{S}_i$ by taking action $a_i \in \mathcal{A}_i$ is expressed as $\Xi_i^{s,s'} = \Pr(s'|s, a_i)$.

4) *Reward for Each Party in PLS*: In time slot t , the immediate reward of $\mathcal{K}, \mathcal{N}$ and $\mathcal{J}$ is respectively defined as

$$\begin{aligned} r_i^t &= U_i(t), \forall i \in \{\mathcal{K}, \mathcal{J}\}, \\ r_{\mathcal{N}}^t &= U_{\mathcal{N}}(t) - \psi \sum_{n \in \mathcal{N}} (R_{nm}^T - R_{min}^T). \end{aligned} \quad (44)$$

where ψ is the penalty for violating the QoS constraint. For conciseness, let $\mathcal{R}_i = \{r_i^t\}_{\forall t}$ denote the reward set of each party $i \in G$.

Note that MDPs for three parties in PLS are inherently coupled [43]. Particularly, i) EVs' decision of activation $x_k^{sw}(t)$ affects LUs' secrecy transmission rate $R_{nm}^S(t)$, which consequently influences LUs's power allocation $P_n^T(t)$ and ii) both LUs' and EVs' unit incentive decision $\mu_L(t)$ and $\mu_E(t)$ in time slot t affect the coalition formation process, which consequently influence JAs's power allocation $P_j^J(t)$. Besides, the hierarchy in these three MDPs, i.e., $\mathcal{M}_{\mathcal{K}}, \mathcal{M}_{\mathcal{N}}$ and $\mathcal{M}_{\mathcal{J}}$, mainly lies in the fact that their decision-making process can be divided into three iterative stages in each time slot, in which $\mathcal{M}_{\mathcal{K}}, \mathcal{M}_{\mathcal{N}}$ and $\mathcal{M}_{\mathcal{J}}$ sequentially generate their decisions in the first, second and third stage based on each MDP's observation, and such process iterates in a back-and-forth manner within each time slot until the convergence.

DRL-Based Solution: Considering the unpredictable channel conditions and the considerably high computational

complexity due to the large action space, we introduce a DRL method based on proximal policy optimization (PPO) to solve the above MDPs, where an actor-critic scheme (AC) [14] is also employed to improve the training efficiency. Specifically, for party $i \in G$, AC includes i) a critic network with parameter ϕ to estimate party i 's state-value $V_{\phi}^i(s) \approx V^i(s)$, where the state value $V^i(s) = \mathbb{E}\{\sum_{t=0}^T \gamma^t r_i^t | s_0 = s, \pi_i\}$ with discount factor γ^t ; ii) an actor network with parameter θ to approximate party i 's optimal mixed strategy Z_i^* ; and iii) a replay buffer to store states, actions and rewards during the training process.

Further notice that for each party in the PLS, its decisions are multi-dimensional, and thus it is very challenging to explore all actions within such a massive space. To circumvent this difficulty, we assign the actions of each single party to multiple virtual agents to increase both exploration probability and training rate. In particular, we set agent $agt_{sw}^{\mathcal{K}}$ and $agt_{\mu}^{\mathcal{K}}$ to generate $x_i^{sw}(t)$ and $\mu_E(t)$ for EVs, N agents $agt_n^{\mathcal{N}}$ to generate $P_n^T(t)$ and $x_{nm}^T(t)$ for each LU $n, \forall n \in \mathcal{N}$, agent $agt_{\mu}^{\mathcal{N}}$ to generate $\mu_L(t)$ for LUs, and agent $agt_p^{\mathcal{J}}$ to generate $P_j^J(t)$ for JAs. We denote the set of all virtual agents as $\mathbb{AGT}$, and enable each agent to adopt the AC with its own actor network, critic network and replay buffer. Note that the mixed strategy of each party $i \in G$ is parameterized by the actor networks of i 's agents. With the update of actor network's parameters of i 's agents (i.e., the policy gradient descent), the mixed strategy of party i can be optimized.

Here, we explain the interactions among all agents in set $\mathbb{AGT}$. In every training step, all agents interact within time slots $t \in [0, T]$, following their corresponding parties' decision sequence defined in the proposed hierarchical game $\mathcal{G}^H$. In each time slot t , after generating each agent's action via its actor network, $NT(t)$ is updated as the next agent's state observation. Note that $NT(t)$ includes the decisions of EVs, LUs and JAs, except for unit incentives which only affect the coalition formation. Then, we ask all three parties to iteratively apply the proposed HCSF algorithm to form the stable coalition partition $\mathcal{CP}^*(t)$, and calculate their rewards $r_i^t, \forall i \in G$. The reward of party i 's agents is the same as i 's reward r_i^t . Afterwards, the replay buffer stores the previous state s_{e-1}^t , the action a_e^t , subsequent state s_e^t and reward r_e^t for each agent $e, \forall e \in \mathbb{AGT}$.

Such hierarchical game guided DRL-based training structure has also been employed in [44] and [45]. Fig. 3 illustrates an overview of our proposed DRL-based solution, consisting of the interactions of all agents, the detailed AC framework of each agent and the update process of both the environment and network parameters (i.e., θ_e of agent e 's actor network and ϕ_e of agent e 's critic network). In this hierarchical game $\mathcal{G}^H$, both the action-generating process and the update process repeat in each training step, and all agents have to run in a back-and-forth manner for achieving long-term performance guarantees.

On top of the above, the network parameters of each agent have to be updated at a certain frequency. The updating process is mainly inspired by the PPO method, namely when the replay buffers of all agents reach the total capacity, we update the actor network and critic network of all agents. For each agent $e \in \mathbb{AGT}$, this update process includes i) the calculation of e 's rewards-to-go, i.e., the discounted rewards $J_e^t = \sum_{t=0}^{T-1} \gamma^t r_t^e$,

⁴In this paper, we delegate the decision-making of LUs, EVs and JAs to three distinct controllers, which can be three specific network device with sufficient computational and sensing capabilities, and their solutions can then be implemented by all corresponding party members for a long-run with long-term performance guarantees. Similar settings are also adopted in [19], [25], and [29].

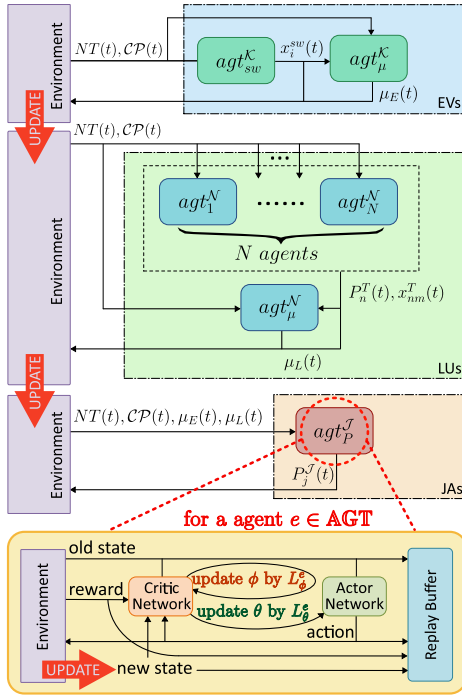


Fig. 3. Overview of the proposed DRL-based solution.

where γ^t stands for the discount factor in time slot t ; ii) the calculation of e 's advantage function $A_t^e = J_t^e - V_\phi^e(s_t^e)$; iii) the calculation of the loss function of e 's actor network: $L_\theta^e = \sum_{t=0}^{T-1} \min(\frac{p_\theta(a_t^e|s_t^e)}{p_{\theta'}(a_t^e|s_t^e)} A_t^e, \text{clip}(\frac{p_\theta(a_t^e|s_t^e)}{p_{\theta'}(a_t^e|s_t^e)}, 1-\epsilon, 1+\epsilon) A_t^e)$, where $p_\theta(a_t^e|s_t^e)$ is the probability of e 's actor network with parameter θ choosing action a_t^e at state s_t^e , θ' is the original parameter of e 's actor network; and iv) the calculation of the loss function of e 's critic network: $L_\phi^e = \sum_{t=0}^{T-1} (V_\phi^e(s_t^e) - J_t^e)^2$. Then, parameters θ and ϕ can be updated for minimizing their corresponding loss function via random gradient descent methods, e.g. Adam optimizer [46]. Algorithm 2 summarizes all detailed steps of the proposed solution.

By running Algorithm 2, all parameters of the actor network and critic network of all agents can be optimized, and thus the mixed strategy equilibrium solution of $\mathcal{G}^H$ (i.e., three parties' optimal mixed strategies parameterized by actor networks of their agents) can be obtained accordingly.

V. SIMULATION RESULTS

In this section, simulations are conducted to evaluate the performance of the proposed DRL-based solution for solving the formulated hierarchical game with dynamic coalitions in the PLS-aware wireless uplink communication system. All simulation results are obtained by taking average over 1000 runs with various system parameters.

A. Simulation Settings

We consider an uplink communication system in a $1000 \text{ m} \times 1000 \text{ m}$ geographic area. There exists $N = 20$ LUs, $M = 5$ BSs, $K = 5$ EVs and $J = 2$ JAs randomly scattered in the target area. The l -leveled transmission power of both LUs and JAs is set within a range of $[0, 20]$ dBm with $l = 10$. The frequency bandwidth W is set as

TABLE II
NEURAL NETWORK SETTINGS

Actor Network		
Linear (input size = D_e^0 , hidden size = 64)		
Linear (input size = 64, hidden size = 64)		
Linear (input size = 64, hidden size = D_e^1)		
Critic Network		
Linear (input size = D_e^0 , hidden size = 64)		
Linear (input size = 64, hidden size = 64)		
Linear (input size = 64, hidden size = 1)		
Agent e	Input Size D_e^0	Output Size D_e^1
agt_{sw}^K	$(N + J) * (M + K) + 2$	2^K
agt_μ^K, agt_μ^N	$(N + J) * (M + K) + 2$	10
$\{agt_n^N\}_{n \in N}$	$(N + J) * (M + K) + 2$	$l * M$
agt_p^J	$(N + J) * (M + K) + 4$	10

$W = 1 \text{ MHz}$ [17], [47]. Furthermore, we assume that $\xi_n = [5, 15]$, $\eta_j = [5, 15]$, $\psi = [0.1, 0.9]$, $c_k = [0.1, 0.5]$, $c_{conf} = [0.3, 0.7]$, $R_{min}^T = 4 \text{ bps/Hz}$, $\mu_L^{max} = 10$ and $\mu_E^{max} = 10$. Note that similar settings have also been employed in [17] and [26]. Besides, the uncertainty of channel gains is modeled with three states, i.e., "Good" with $E\{g_{nm}(t)\} = 0.6$, "Normal" with $E\{g_{nm}(t)\} = 0.4$ and "Bad" with $E\{g_{nm}(t)\} = 0.2$, while their switching probabilities are set following [10]. Besides, the structure of each agent's actor network and critic network, and hyper-parameter settings for the proposed DRL-based solution is presented in Table II. Note that some parameters may vary for different evaluation purposes.

We compare our proposed DRL-based solution with the following benchmarks:

- *EFJ-DRL*: EV's friendly jamming based on DRL, in which JAs always and unilaterally choose to aid EVs in eavesdropping by jamming the uplink communication links, while the other decisions are made by the DRL-based algorithm [48].
- *LFJ-DRL*: LUs' friendly jamming based on DRL, in which JAs always and unilaterally choose to aid LUs in improving secrecy transmission rates by jamming the eavesdropping links, while the other decisions are made by the DRL-based algorithm [9].
- *EFJ-PSO*: EV's friendly jamming based on particle swarm optimization algorithm (PSO), in which JAs always and unilaterally choose to aid EVs in eavesdropping by jamming the uplink communication links, while the other decisions are made by PSO-based algorithm [49].
- *LFJ-PSO*: LUs' friendly jamming based on PSO, in which JAs always and unilaterally choose to aid LUs in improving secrecy transmission rates by jamming the eavesdropping links, while the other decisions are made by PSO-based algorithm [49].
- *R^T -Max*: LUs greedily maximize $\{R_{nm}^T(t)\}_{n \in N}$ in each time slot when determining the transmission power allocation $\{P_n^T(t)\}_{n \in N}$ and BS association $\{x_{nm}^T(t)\}_{n \in N}$.
- *R^S -Max*: LUs greedily maximize $\{R_{nm}^S(t)\}_{n \in N}$ in each time slot when determining the transmission power allocation $\{P_n^T(t)\}_{n \in N}$ and BS association $\{x_{nm}^T(t)\}_{n \in N}$ [22].

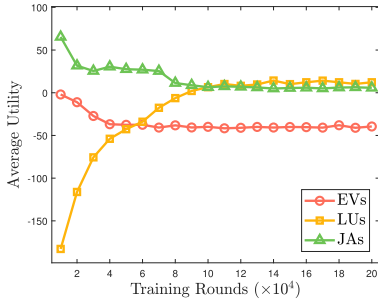
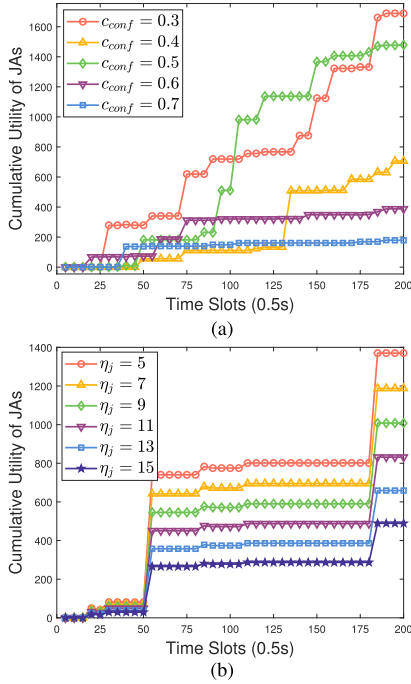


Fig. 4. Convergence of the proposed DRL-based solution.

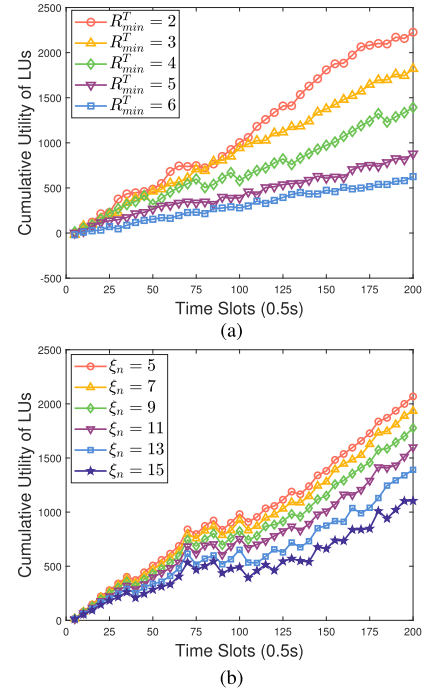
Fig. 5. JAs' performance with different c_{conf} and η_j .

- $R^E\text{-Max}$: EVs greedily maximize $\{R_{nk}^E(t)\}_{\forall k}$ in each time slot when determining the activation of each EV in different geographical locations, i.e., $\{x_k^{sw}(t)\}_{\forall k}$ [50].
- $V_{\mathcal{J}}\text{-Max}$: JAs greedily maximize their total incentive $V_{\mathcal{J}}(t)$ in each time slot when deciding the jamming power allocation $\{P_j^J(t)\}_{\forall j}$ [6].

B. Performance Evaluation

Fig. 4 examines the convergence of the proposed DRL-based solution. It is shown that, the utilities of all three parties converge very quickly, which implies that the mixed strategies (Z_K^*, Z_N^*, Z_J^*) can quickly converge to the equilibrium of the formulated hierarchical game $\mathcal{G}^H$. This is owing to the adjacency matrix and virtual agents designed for reducing the complexity of both state and action spaces for improving the training efficiency, and the adoption of PPO in the strategy exploration for accelerating the convergence. Such convergence results well match the theoretical analyses provided in Section IV, and verify the feasibility of our proposed solution.

Fig 5 depicts the performance of JAs using the proposed DRL-based solution, and shows how JAs' cumulative utility is affected by the configuration cost c_{conf} and the unit jamming power cost η_j . In Fig 5(a), the curve of JA's cumulative utility

Fig. 6. LUs' performance with different R_{min}^T and ξ_n .

exhibits a step-wise pattern, where each upward segment represents a change in the coalition partition of all three parties. It can be observed that, as c_{conf} increases, the number of upward segments in the curve decreases. This is because when c_{conf} is sufficiently large, JAs become less willing to change the coalition partition and may even choose to not assist either LUs or EVs. In Fig 5(b), we can observe that the slope of the curve for JA's cumulative utility decreases more gradually as η_j increases, compared to that in Fig 5(a). This is because the proposed DRL-based solution trains JAs to achieve a stable and optimal power allocation strategy, such that the jamming power allocations under different settings of η_j are similar. In other words, the increment in η_j only leads to a slight increase in the total power cost, resulting in a gradual decrement of JA's cumulative utility, and has little impact on the coalition formation process.

Fig. 6 evaluates the performance of LUs using the proposed DRL-based solution, and demonstrates the impact of the minimum transmission rate R_{min}^T and the unit transmission power cost ξ_n on LUs' cumulative utility. In Fig. 6(a), it can be observed that LUs' performance decreases with increasing R_{min}^T . This is because when R_{min}^T increases, LUs need to increase their transmission powers and choose the target BSs that are geographically closer but more vulnerable to eavesdropping to satisfy the QoS constraint. In Fig. 6(b), it is obvious that LUs' performance decreases with increasing ξ_n . The reason for this is that if ξ_n increases, LUs will reduce their transmission powers to lower the total power consumption. Furthermore, it can be observed in Fig. 6(a) that R_{min}^T varying from 2bps/Hz to 6bps/Hz results in a decline of over 400% in LUs' utility, but such a dramatic decline does not occur when ξ_n increases in Fig. 6(b). The reason is that R_{min}^T affects both the BS association and power allocation of LUs, whereas ξ_n does not actually influence the BS association.

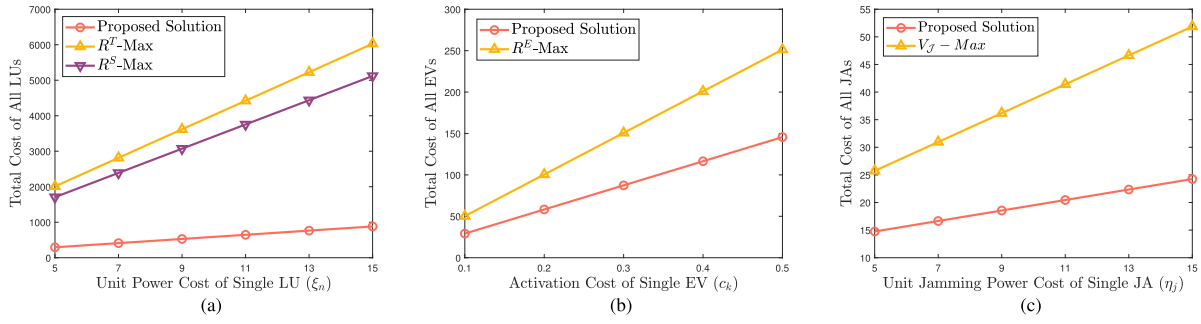


Fig. 7. Performance comparison among different approaches by varying ξ_n , c_k and η_j .

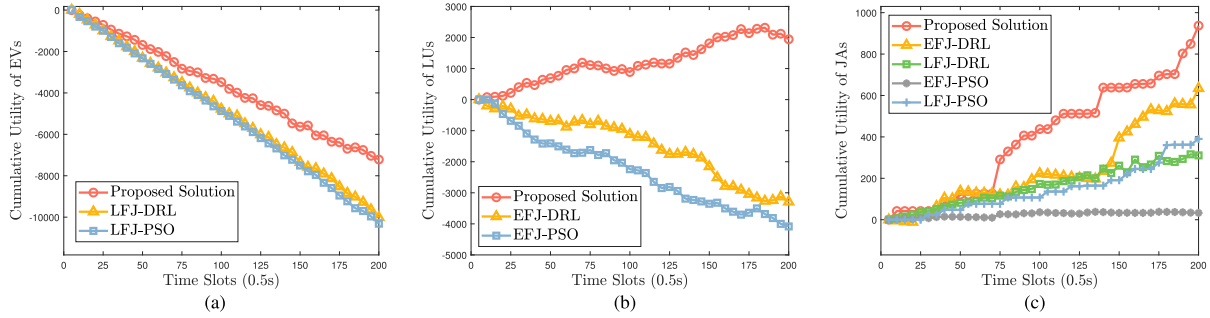


Fig. 8. Performance comparison in terms of EVs', LUs' and JAs' cumulative utilities.

Fig. 7 presents a comparison of the performance of the proposed DRL-based solution with R^T -Max, R^S -Max, R^E -Max, and V_J -Max, by varying the costs of LUs' unit transmission power ξ_n , EVs' activation c_k , and JAs' unit jamming power η_j . As shown in Fig. 7(a), the proposed solution achieves significantly lower total cost for all LUs compared to R^T -Max and R^S -Max. Similarly, Fig. 7(b) demonstrates that the proposed solution outperforms R^E -Max in terms of the total cost for all EVs, while in Fig. 7(c), the total cost for all JAs is considerably lower than that obtained by V_J -Max. These results indicate that the proposed DRL-based solution is superior in minimizing the total cost for all parties over all benchmarks. This is because that existing approaches (i.e., R^T -Max, R^S -Max, R^E -Max, and V_J -Max) solely maximize their respective physical quantities (i.e., $R_{nm}^T(t)$, $R_{nm}^S(t)$, $R_{nk}^E(t)$, and $V_J(t)$) while neglecting the costs incurred by higher $P_n^T(t)$ of LUs, number of activated devices of EVs (i.e., $\sum_{\forall k \in \mathcal{K}} x_k^{sw}(t)$), and $P_j^J(t)$ of JAs, respectively. In contrast, the proposed solution considers all possible factors that impact the performance of all parties by formulating their utility functions more comprehensively and well addresses the dynamic strategic interactions among them.

Fig. 8 demonstrates the superiority of the proposed solution with dynamic coalition in terms of EVs', LUs' and JAs' cumulative utilities. It can be observed that i) in Fig. 8(a), the proposed solution outperforms LfJ-DRL and LfJ-PSO in terms of EVs' utility; ii) in Fig. 8(b), the proposed solution outperforms EFJ-DRL and EFJ-PSO in terms of LUs' utility; and iii) in Fig. 8(c), the proposed solution outperforms the others in terms of JAs' utility. This is because the proposed solution allows JAs to dynamically form coalitions with either LUs or EVs based on the time-varying system states, which helps them earn more rewards for their services in either secrecy transmission or eavesdropping. In other words, unlike

the fixed relationships among EVs, LUs and JAs imposed in either EVs' Friendly Jamming or LUs' Friendly Jamming based on either DRL or PSO, the dynamic trilateral coalitions considered by the proposed solution provides higher flexibility and adaptability to the practical PLS-aware wireless networks. Moreover, it is shown in Fig. 8 that the DRL-based solutions outperform the PSO-based solutions. This is because DRL-based solutions have stronger exploration capabilities than PSO-based ones in dynamic wireless communications, reducing the possibility of falling into the local optima.

VI. CONCLUSION

In this paper, we propose a novel hierarchical game framework for PLS-aware wireless communications. To be more specific, with the aim of modeling the strategic interactions among EVs, LUs and JAs under the dynamic nature of the wireless system and their potential coalition relationships. We then formulate a hierarchical game integrating with a trilateral CFG. By analyzing stability conditions of the underlying trilateral coalitions, we propose an HCSF algorithm for stable coalition partition in each time slot. Then, after analyzing the existence of the overall hierarchical game's equilibrium, we propose a DRL-based solution which can converge to a mixed-strategy equilibrium for the formulated hierarchical game with long-term performance guarantees. Simulation results verify the feasibility of the proposed solution and demonstrate its superiority over its counterparts. In increasing the cumulative utilities of EVs, LUs and JAs.

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